

Analysis of Algorithms (BDSF25M/A)

Date: 06-May-2026

Quiz 05

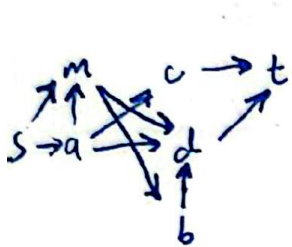
Time: 25 min

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Grading	CLO-1	CLO-2	CLO-3	CLO-4
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Run BFS on the following graph

(start from s, then on requirement scan vertices from top to bottom and edges from left to right)



Vertex	Adj. Vertices	color	d (distance/edges)	p (predecessor)
a	m c d	gray White	∞ 1	None S
b	d	gray White	∞ 2	None m
c	t	gray White	∞ 2	None a
d	t	gray White	∞ 2	None a
t		gray White	∞ 3	None c
m	b d	gray White	∞ 1	None s
s	a m	White	∞ 0	None

Starting from s node s colour = gray, $d[s] = 0$, $P[s] = \text{None}$, enqueue(s)

d	0	1	1
Q =	s	a	m
P	none	s	s

dequeue(s): enqueue neighbours of s (a,m) in order
 $d[a], d[m] = 1$, $P[a], P[m] = s$, colour[a], colour[m] = gray

visits a:	1	2	2
Q =	m	c	d
	s	a	a

colour[a] = black

visits m:

Q =

c	d	b	
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colour[m] = black

$$Q = \begin{array}{c|c} 2 & 3 \\ \hline b & t \\ \hline m & c \end{array}$$

visit c: $d[c] = \text{black}$, visit d: + visited already $c[d] = \text{black}$
 visit b: $d[b] = \text{black}$, visit t: $d[t] = \text{black}$

Run DFS on the following graph

(when required scan vertices from top to bottom and edges from left to right)

Vertex	Adj. Vertices	color	d (discovery time)	f (finish time)	p (predecessor)
m	b d	White	1	8	None
b	d	White	2	7	None m
a	m c d	White	9	12	None
c	t	White	10	11	None a
s	a m	White	13	14	None
d	t	White	3	6	None b
t		White	4	5	None d

Time	discover	Colour		
1	m	gray	m b d t t d b m	8
2	b	gray		7
3	d	gray		6
4	t	gray		5
5	finish t	Black		4
6	finish d	Black		3
7	finish b	Black		2
8	finish m	Black		1
9	a	gray	s s a c a	14
10	c	gray		13
11	finish c	Black		12
12	finish a	Black		11
13	s	gray		10
14	finish s	Black		9

m → b → d → t

a → c

s